Royal Devon University Healthcare NHS Foundation Trust

Acute Scrotal Pain Pathway – Clinical Guidance (Northern)

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Are there any previous pieces of guidance relating to this topic?	Adapted from equivalent pathway in Eastern
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SUMMARY

This document provides guidance for the diagnosis, management and follow up of acute scrotal pain. It is based on the pathway in use at Eastern and adapted for use in Northern and to incorporate recent publications from NCEPOD and GIRFT. There is a summary flow chart (Appendix 1) which can be used as a quick-reference guide and will be displayed in relevant clinical areas.

KEY POINTS

- Acute scrotal pain (ASP) should raise suspicion of a possible testicular torsion. Testicular torsion is a surgical emergency and suspicion warrants urgent scrotal exploration.
- Any child/young male presenting with abdominal pain should have a testicular examination performed.
- In a testicular torsion the blood supply to the testis is cut off due to a twist of the testis on its axis. The chance of the testis surviving decreases the longer it is left, with high likelihood of the testis dying after 6 hours.
- Recent guidelines from GIRFT and NCEPOD emphasise the important of prompt assessment by a surgical decision maker and urgent surgery if appropriate.
- All referrals for acute scrotal pain should be made via bleep to the on call surgical registrar (bleep 900)
- Intra-operative management depends on pathology found and the decision-making tool from the BURST trial can aid in this process.
- All patients who undergo orchidopexy for confirmed testicular torsion should be followed up.

CONTENTS

CONTENTS

1. INTRODUCTION 4
2. BACKGROUND 4
3. DEFINITIONS 4
4. ASSESSMENT & MANAGEMENT OF ACUTE SCROTAL PAIN 4
4.1 Referral 4
4.2 Assessment [1,2,3] 5
4.3 Management 5
4.4 Follow up 7
5. MONITORING COMPLIANCE WITH THIS GUIDELINE 8
6. REFERENCES 9
Appendix 1 – Acute Scrotal Pain Pathway 10
Appendix 2 – Decision making flow chart [1] 11

1. INTRODUCTION

1.1 This clinical guideline is designed to provide advice, guidance and direction to staff whilst leaving room for professional judgement, and adaptation, to fit individual circumstances.

1.2 This guideline applies to patients of all ages presenting with acute scrotal pain in the Northern site.

2. BACKGROUND

2.1 This is a new guideline to improve care for patients with acute scrotal pain. It aims to provide clarity about who to contact and to aid with decision making.

2.2 Acute scrotal pain (ASP) should raise suspicion of a possible testicular torsion. Testicular torsion is a surgical emergency and suspicion warrants urgent scrotal exploration. In a testicular torsion the blood supply to the testis is cut off due to a twist of the testis on its axis. The chance of the testis surviving decreases the longer it is left, with high likelihood of the testis dying after 6 hours.

Whilst there is a bimodal distribution of torsion (neonate and more commonly early adolescent boys) torsion needs to be ruled out at any age.

3. DEFINITIONS

3.1 Orchidopexy – surgical fixation of the testis within the scrotum to prevent further episodes of torsion

3.2 Orchidectomy – surgical removal of the testis

4. ASSESSMENT & MANAGEMENT OF ACUTE SCROTAL PAIN

4.1 **Referral**

Patients with ASP may be referred from their GP or from the Emergency Department (ED). Referrals from a GP should be advised to attend KGV AAA (over 18) or ED (under 18) as soon as possible and stay nil by mouth. On arrival the ED triage team or AAA team should contact the surgical registrar on bleep 900 immediately.

For patients self-presenting to ED with less than 48 hour history of acute unilateral testicular pain the triage team should make a direct referral to the surgical registrar on bleep 900 as soon as possible.

4.2 **Assessment** [1,2,3]

The surgical registrar should see the patient within 1 hour of arrival. If they are unavailable they should send a junior doctor (ideally bleep 333) to assess the patient and report back to them.

History

Testicular torsion commonly presents with acute unilateral scrotal or abdominal pain which may or may not be associated with nausea or vomiting. The history should also include time of onset, any previous episodes (to suggest intermittent torsion), any history of trauma, urinary symptoms and a sexual history if the patient is sexually active. A TWIST score should be calculated.

Examination

Examine the abdomen, groin and scrotum including checking the cremasteric reflex.

Investigations

Urine dip. Send for STI screen if patient is sexually active. Bloods are not usually helpful.

Imaging should only be considered if there is a more likely alternative pathology that would change management or if the pain has been present for more than 48 hours. It should not delay surgery if surgery is indicated.

Transfer to a tertiary centre (Bristol Children's Hospital) is not usually indicated and should not occur without a consultant to consultant discussion. Telephone advice can be sought if there is diagnostic uncertainty.

4.3 **Management**

All patients with suspected testicular torsion need surgical exploration within 1 hour of deciding to operate. Fasting status is not relevant. Scrotal exploration can be performed by either general surgery or urology teams – both specialties mandate proficiency for CCT. If there is not an available theatre within an appropriate time there should be consideration for opening another theatre for timely surgery. Liaise with the theatre co-ordinator on bleep 256 to discuss this.

Consent

Consent the patient for orchidectomy, decreased fertility, infection (in the testicle or the wound), haematoma that may require drainage, contralateral testicular injury and anaesthetic complications including the risks of not being adequately fasted.

Recommend or provide a copy of the BAUS information leaflet: [4]
https://www.baus.org.uk/_userfiles/pages/files/Patients/Leaflets/Torsion%20of%20testis.pdf

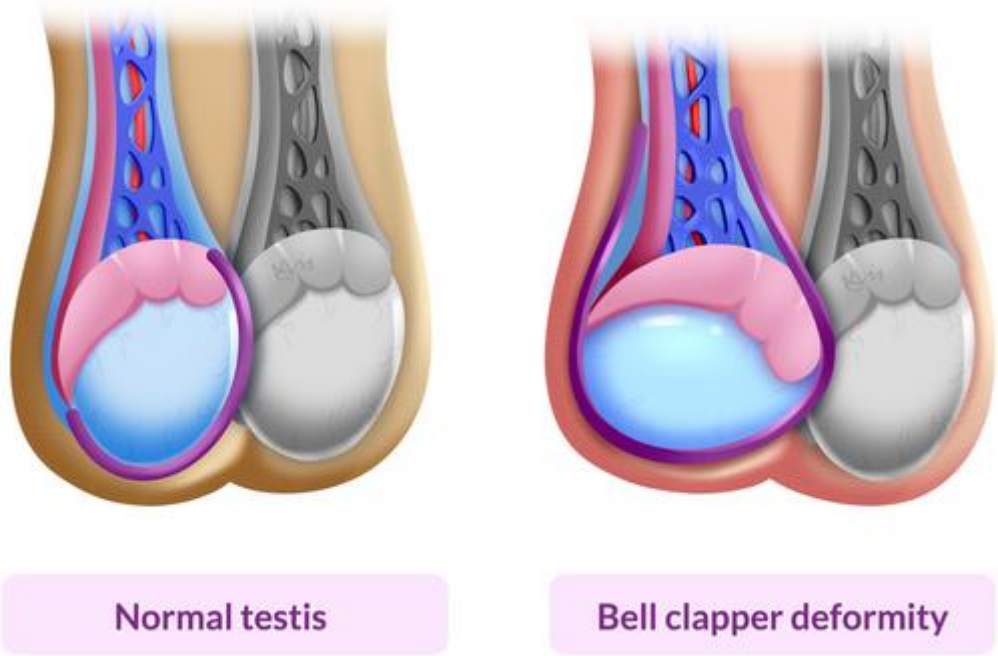
Operative findings and decision making

A recent consensus document [1] has made the following recommendations for management of testes according to the operative findings:

Testicular torsion

Untwist testis, wrap in warm saline soaked gauze & fix contralateral side. If testis is not viable proceed to orchidectomy. If viability is uncertain then fix – do not perform orchidectomy.

If there is no evidence of torsion, the presence or absence of a bell-clapper abnormality determines subsequent management:



No torsion, no other pathology seen
If no torsion found and no other abnormalities are seen to account for presentation do not fix testis.

If no torsion found and no other abnormalities are seen, but bell-clapper abnormality is present then fix the testis and explore contralateral side and fix if bell-clapper abnormality is seen.

No evidence of testicular torsion but other abnormality seen to account for presentation
Examples of other pathology include epididymo-orchitis, torted testicular appendage, mumps orchitis and testicular tumour. Torted appendage can be excised.

If bell-clapper abnormality is present then fix the testis and explore contralateral side (unless there is epididymo-orchitis) and fix if bell-clapper abnormality is seen.

See Appendix 2 for detailed decision-making flow chart

For children either a Dartos pouch or suture fixation can be performed (depending on surgeon experience). Adults should undergo a suture fixation – either 3 or 4 point fixation is acceptable. Either non-absorbable or prolonged-absorption sutures can be used.

Patients with epididymo-orchitis should be treated according to the trusts antibiotic guidelines (On Hub or Microguide app). Patients with risk factors for sexually-transmitted infections should be advised to attend the sexual health clinic. All patients should see their GP within 2-4 weeks to assess resolution and need for ultrasound or other investigations. All patients must be advised to return to ED for re-assessment at once if the condition worsens or they have any concerns.

4.4 Follow up

All patients with confirmed torsion that undergo fixation should be followed up at 6 months to assess for delayed atrophy.[2]

Patients under age 18 – order a face to face follow up with “Paediatrics – Surgery”
Patients age 18 and over – order a face to face follow up with the responsible Urologist (who is on call when the patient is admitted)

Patients who have an orchidectomy should be given patient-initiated follow up (PIFU) in case they require further information or support. Prosthetics can be discussed in post-pubertal patients.

Patients who do not have confirmed torsion do not require a follow up.

Discharge advice should include details about wound care and sutures, follow up arrangements and who to contact if they have concerns. Patients should also be given advice regarding prosthetics – these cannot be offered until puberty is complete. If the patient is still on PIFU they can request an appointment, otherwise their GP can refer them to either paediatric surgery (if under 18) or urology (if over 18).

5. MONITORING COMPLIANCE WITH THIS GUIDELINE

5.1 Compliance with this guideline will be monitored with regular audit – particularly regarding time to review by surgical decision maker and time to theatre.

6. REFERENCES

1. Keiran D. Clement, Alexander Light, Aqua Asif, Vinson Wai-Shun Chan, Sinan Khadhouri et al. A BURST-BAUS consensus document for best practice in the conduct of scrotal exploration for suspected testicular torsion: the Finding Consensus for Orchidopexy in Torsion (FIX-IT) study *BJU International 2022*; 130 (5): 662-670. Available at: - https://bjui-journals.onlinelibrary.wiley.com/doi/full/10.1111/bju.15818 (Accessed 16th July 2024)
2. Getting It Right First Time (NHS Impact). *GIRFT Children and Young People: Testicular torsion pathway.* Available at https://gettingitrightfirsttime.co.uk/wp-content/uploads/2024/02/Paediatric-testicular-torsion-pathway-guide-FINAL-V1-February-2024.pdf (Accessed 16th July 2024)
3. British Association of Paediatric Surgeons, British Association of Paediatric Urologists. *Commissioning guide: Management of Paediatric Torsion*. Available at: https://www.england.nhs.uk/mids-east/wp-content/uploads/sites/7/2018/03/torsion-commissioning-guide.pdf (Accessed 16th July 2024)
4. British Association of Urological Surgeons. *Scrotal Exploration for Suspected Torsion of the Testis.* Available at: https://www.baus.org.uk/_userfiles/pages/files/Patients/Leaflets/Torsion%20of%20testis.pdf (Accessed 16th July 2024)

Appendix 1 – Acute Scrotal Pain Pathway

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Acute Scrotal Pain Pathway – Northern

Pathway applies to all patients regardless of age or time of day

GP referral

Advise GP to send patient to ED/AAA urgently and stay NBM

Inform surgical decision maker (bleep 900) of expected patient *(if referral taken by a different member of the team)*

On arrival ED triage team or AAA ward clerk should bleep 900 immediately for review

ED referral

Triage team to refer directly to bleep 900 if patient has acute, unilateral scrotal pain of less than 48 hour history

If the patient does not fit these criteria they should be triaged and assessed by ED in the normal way

Surgical review

Patient should be seen by a surgical decision maker (on call surgical registrar) within 1 hour of arrival to ED

If unavailable they should arrange for bleep 333 to see the patient instead and report back

Perform history, examination and TWIST score** - **note that patients with learning disabilities or neurodivergent conditions may present atypically and under-report pain – have a higher index of suspicion in these cases**

No need for ultrasound unless an alternative pathology suspected that would change management (e.g. a tumour)

Suspected torsion requires urgent surgical exploration under general surgery

Urology advice will be available within normal hours (0800 – 1700) Monday to Friday if required

Book – SUR2 order on EPIC, bleep 256 (theatre coordinator) and 508 (anaesthetics)

Consent and mark*

Scrotal Exploration

Should happen within 1 hour of deciding to operate

Midline raphe or transverse incision can be used

Proceed depending on intra-operative findings *

TWIST Score**	
Testicular swelling	2
Hard testis	2
High-riding testis	1
Nausea/vomiting	1
Absent cremasteric reflex	1

Score ≥ 5 – mandates exploration

Score 0-4 does not exclude torsion

Follow up

No torsion/fixation – no follow up required

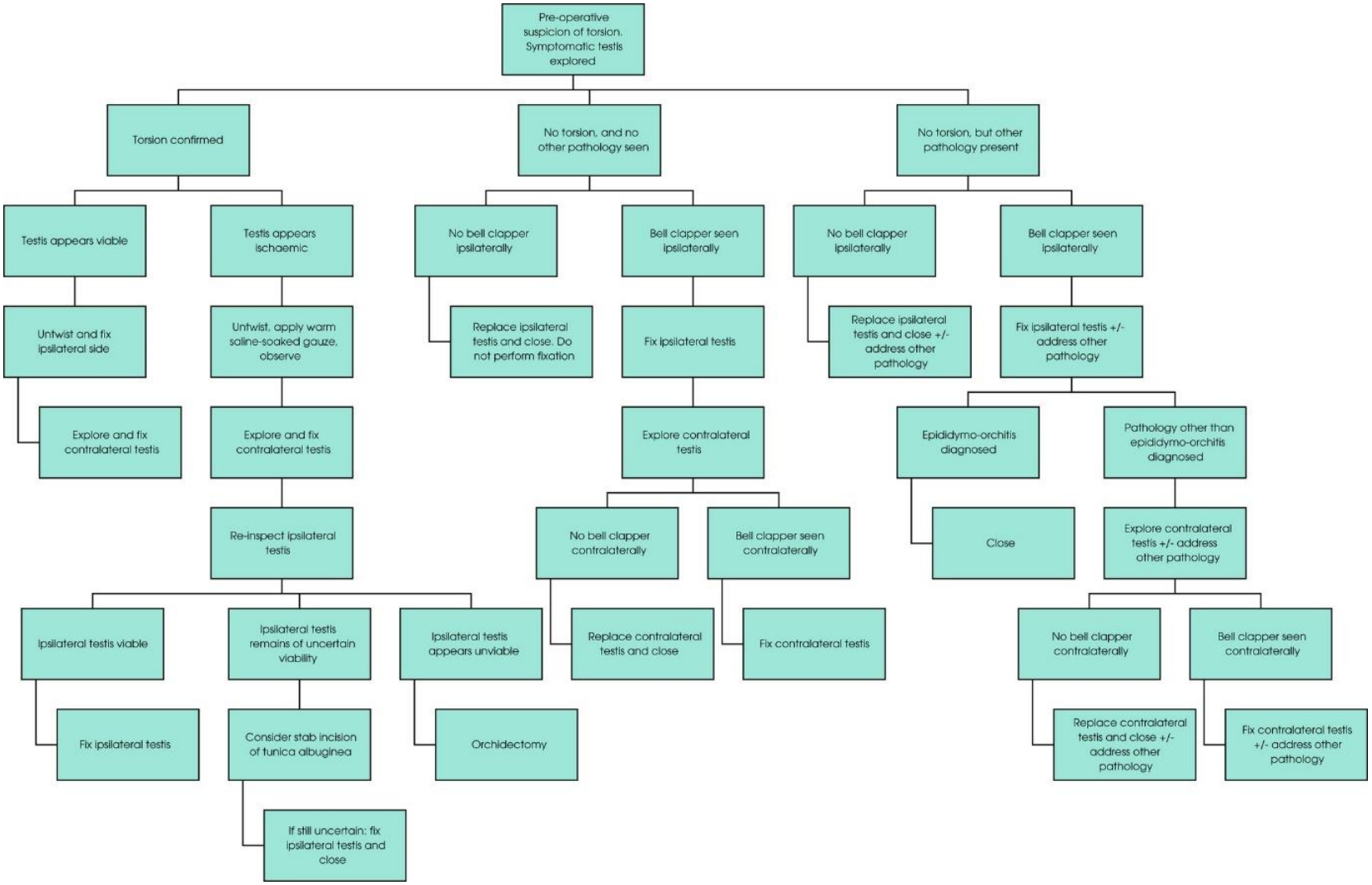
Torsion with orchidopexy – Face to face in 6 months (see below)

Torsion with orchidectomy – Patient-initiated follow up (see below)

Under 18 – Book appointment with "Paediatrics – Surgery" in 6 months (any consultant)

Over 18 – Book appointment with Urologist responsible for patient in 6 months

Provide follow up information*

* See full pathway document for further information

Appendix 2 – Decision making flow chart [1]